Design, Embedded Implementation, and Identification of a Minimal-Sensing Indoor Blimp

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***Resumen*—This paper presents the design, embedded implementation, physical modeling, and experimental identification of a miniature VTOL (Vertical Takeoff and Landing) blimp based on a 50-inch Mylar envelope operating under quasi-neutral buoyancy conditions. The vehicle employs two counter-rotating coreless DC motors mounted on a servo-actuated balsa shaft that provides 180 degrees thrust vectoring, allowing the same propulsion system to generate both vertical and horizontal forces. A lightweight sensing architecture, consisting of a VL53L1X Time-of-Flight sensor for altitude measurement and an MPU6050 inertial measurement unit for heading estimation, is integrated with an ESP32-C6 microcontroller that manages sensing, actuation, and wireless communication. Physics-based dynamic models are derived for the vertical and yaw motions, accounting for buoyancy, gravity, aerodynamic drag, and actuator dynamics. Model parameters are experimentally identified from flight data acquired under quasi-neutral buoyancy conditions. The identified yaw model incorporates asymmetric actuator gains together with a nonlinear quadratic aerodynamic damping term, accurately reproducing the measured vehicle dynamics over the operating range. Beyond the dynamic modeling, the paper provides the complete engineering design of a reproducible indoor blimp platform, including its mechanical structure, propulsion system, power electronics, and sensing architecture.**

***Index Terms*—VTOL blimp, indoor blimp, quasi-neutral buoyancy, thrust vectoring, dynamic modeling, system identification, aerial robotics.**

I. Introduction

Lighter-than-air (LTA) vehicles have long attracted interest as aerial robotic platforms because buoyancy provides most of the lift, allowing propulsion to be devoted primarily to maneuvering rather than weight support. Compared with multirotor aerial vehicles, indoor blimps offer significantly longer endurance, lower energy consumption, quieter operation, and inherently safer interaction with humans due to their low operating speeds and compliant envelopes. These characteristics make them attractive for persistent indoor inspection, environmental monitoring, education, warehouse automation, and human–robot interaction.

Research on autonomous blimps experienced substantial growth during the late 1990s and early 2000s, when several groups established the theoretical and experimental foundations for autonomous lighter-than-air vehicles. Among the most influential efforts, the LAAS/CNRS Autonomous Blimp Project developed one of the first complete experimental platforms integrating vehicle design, dynamic modeling, flight control, localization, and terrain mapping. Their work demonstrated that autonomous blimps constitute a challenging control problem because of their nonlinear aerodynamics, low damping, actuator limitations, and slowly varying buoyancy, while also highlighting the advantages of decomposing the vehicle dynamics into nearly independent longitudinal and lateral motions [1], [2]. These pioneering contributions established many of the modeling and control methodologies that continue to influence current research.

Following a period of comparatively reduced activity, autonomous indoor blimps have experienced a remarkable resurgence during the last few years. This renewed interest has been enabled by lightweight embedded processors, high-energy-density batteries, inexpensive MEMS sensors, and miniature propulsion systems, which have considerably reduced the size and cost of experimental platforms. The Georgia Tech Miniature Autonomous Blimp (GT-MAB) represents one of the most representative examples of this new generation of platforms, addressing mathematical modeling, system identification, switched control, autonomous waypoint navigation, and human–robot interaction using miniature indoor airships [3], [4].

More recently, the research community has expanded beyond conventional PID-based control toward increasingly sophisticated approaches. Reinforcement learning has been investigated for autonomous flight and trajectory tracking, demonstrating promising robustness against aerodynamic uncertainties and slowly varying buoyancy conditions [5], [6]. In parallel, new robotic platforms have explored novel vehicle

configurations. RGBlimp investigated the design, modeling, and aerodynamic characterization of a robotic gliding blimp [7], whereas MochiSwarm demonstrated the feasibility of cooperative micro-blimp swarms operating in realistic environments [8]. The recently proposed Spinning Blimp introduced a minimalist aerial vehicle that exploits rotational dynamics as its primary locomotion mechanism [9]. At the same time, increasingly sophisticated control strategies have been developed, including adaptive output-feedback trajectory tracking [10] and swing-reduction control for underactuated indoor blimps [11]. Complementing these efforts, recent work on system identification has highlighted the importance of accurate plant modeling as the foundation for high-performance feedback control [12]. The recent appearance of robotics competitions focused on autonomous lighter-than-air vehicles has further stimulated interest in lightweight experimental platforms and embedded flight systems.

Despite these advances, most contemporary research platforms rely on relatively sophisticated sensing architectures involving onboard cameras, visual-inertial odometry, LiDAR sensors, ultra-wideband localization, motion-capture systems, or embedded computers capable of executing computationally intensive perception algorithms. Although these solutions enable fully autonomous navigation, they also increase payload, power consumption, hardware complexity, and implementation cost. Comparatively less attention has been devoted to the development of lightweight experimental platforms that intentionally employ minimal onboard sensing while remaining suitable for system identification, controller development, and robotics education.

This work addresses that gap by presenting the design, physical modeling, experimental identification, and embedded implementation of a miniature indoor VTOL blimp based on a deliberately minimal sensing architecture. The proposed vehicle employs only a VL53L1X Time-of-Flight sensor for altitude measurement and a low-cost MPU6050 inertial measurement unit for heading estimation, both managed by an ESP32-C6 microcontroller. The propulsion system consists of two counter-rotating coreless DC motors mounted on a servo-actuated vectored-thrust mechanism, enabling the same propulsion system to generate both vertical and horizontal forces while maintaining a compact mechanical design. The system can be replicated using the implementation resources provided with this work, including the electronic wiring diagram, the CAD design of the gondola, the MATLAB-based graphical user interface, and the ESP32 firmware. All these resources are publicly available at https://github.com/jairc28/VTOL-BLIMP.

Physics-based dynamic models are derived for both the vertical and yaw dynamics. Their parameters are experimentally identified from flight data acquired under quasi-neutral buoyancy conditions, resulting in compact models suitable for subsequent controller design. During the experimental implementation, the propulsion system exhibited pronounced dead-zone nonlinearities at low PWM commands, preventing reliable closed-loop validation around the hovering operating point. The identified models constitute the basis for future work on embedded feedback control, as well as some possible implementation improvements.

The remainder of this paper is organized as follows. Section II presents the mechanical, electronic, and sensing architecture of the proposed platform. Section III develops the dynamic modeling and experimental parameter identification. Section IV includes some practical recommendations when implementing the system and experimentally characterizing the platform. Finally, Section V summarizes the main conclusions and outlines future research directions.

II. System Description

We were interested in building a VTOL blimp platform with quasi-neutral buoyancy, in order to minimize the energy needed for its motion. The platform should be able to translate on the vertical and longitudinal axis, as well as rotate along the vertical axis, which enables it to move on the three-dimensional space. In order to control at some point the blimp and to maintain the minimal-sensing characteristic, we need to measure blimp altitude and yaw motions. The platform must be completely autonomous; therefore, all the electronics required for its operation, as well as the power source, must be carried on board. Fig. 1, on the left, shows the complete VTOL blimp platform implemented, while Fig. 1, on the right, presents the electronic gondola and vectored-thrust propulsion system. The following subsections describe the main engineering decisions underlying the platform design, including the aerostatic configuration, mechanical structure, propulsion system, electronics, sensing, and power architecture.

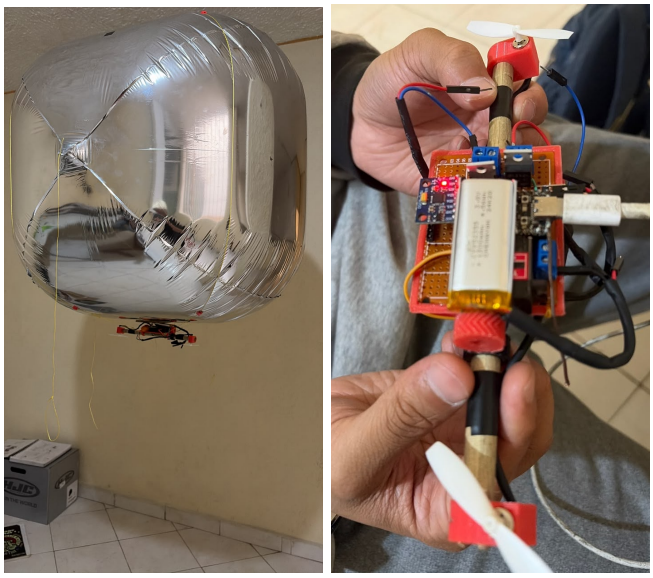


Figura 1. VTOL blimp implemented (left), and electronic gondola (right).

II-A. Aerostat and Quasi-Neutral Buoyancy

The lifting envelope consists of a 50-inch (approximately 1.27 m diameter) helium-filled Mylar balloon. This diameter was selected to satisfy the quasi-neutral buoyancy requirement, while providing sufficient payload capacity for the embedded

control platform. The net buoyant force generated by a helium volume, V, is

$$F_b = (\rho_{\text{air}} - \rho_{\text{He}})\, gV - W_{\text{film}}, \tag{1}$$

where $\rho_{\text{air}} \approx 1.06\ \text{kg/m}^3$, and $\rho_{\text{He}} \approx 0.16\text{kg/m}^3$ correspond to the atmospheric conditions in Bogotá (approximately 2600 m above sea level), and W_{film} denotes the weight of the balloon envelope. Under these conditions, helium provides a net lifting capacity of approximately 0.9g/L. After accounting for the envelope mass, a 50-inch balloon provides sufficient lift to support the approximately 100g gondola, while preserving a small buoyancy margin for trimming.

Fine adjustment to the quasi-neutral operating point is achieved by attaching small pieces of modeling clay to the gondola. This ballast enables rapid and continuous mass adjustment during static calibration, compensating for the residual buoyancy imbalance, $F_b - W_{\text{total}}$. Unlike fixed ballast, modeling clay can be easily added or removed to accommodate buoyancy variations caused by ambient temperature changes and the gradual helium permeation through the Mylar envelope.

II-B. Mechanical Design of the Gondola and Suspension System

The gondola is attached to the balloon through a rigid suspension system based on an inelastic thread. Nylon and other elastic materials were intentionally avoided because their compliance would allow pendulum-like oscillations induced by the propulsion system, introducing unmodeled dynamics that could degrade yaw behavior. The inelastic suspension maintains a constant attachment length under mechanical loading and temperature variations, preserving a fixed relative position between the gondola and the balloon envelope.

The 3D-printed chassis incorporates integrated cable guides that constrain the suspension line along a predefined path and an adjustable tension knot that allows the gondola position to be accurately trimmed. Positioning the gondola at the balloon equator minimizes disturbance-induced moments, while keeping the center of mass below the center of buoyancy, thereby providing passive pendulum stability through a restoring moment under small angular displacements.

To minimize structural mass while maintaining adequate rigidity, the chassis was fabricated using fused-filament 3D printing with a $10\,\%$ infill density. The structure adopts a closed, triangulated ring that supports the transverse balsa shaft carrying the propulsion system. This geometry provides high structural stiffness and prevents deformation caused by the radial forces transmitted through the printed gears. Balsa wood was selected for the motor shaft because of its excellent stiffness-to-weight ratio, allowing the assembly to remain lightweight, while avoiding shaft deflection under the torque generated by the SG90 servo motor.

II-C. Propulsion and Thrust Vectoring System

The propulsion system consists of two counter-rotating coreless DC motors, equipped with propellers and, mounted at the ends of a transverse balsa shaft. Each propeller is matched to its corresponding direction of rotation, allowing the reaction torques generated by the motors to cancel each other. This configuration prevents undesired rotations caused by net motor torque and improves directional stability during both hovering and forward flight. Coreless motors were selected because of their high thrust-to-weight ratio and fast dynamic response, both of which are essential under the stringent mass constraints of the platform.

Thrust vectoring is achieved by rotating the entire propulsion assembly through a 180° range using an SG90 micro servo. This mechanism continuously redirects the combined thrust vector from the vertical direction $\hat{z}$, providing lift for ascent and descent, to the horizontal direction $\hat{x}$, enabling forward motion and yaw maneuvers. The SG90 was selected for its low mass, low power consumption, and sufficient torque to actuate the balanced propulsion assembly.

By varying the vectoring angle, the total thrust is decomposed into vertical and horizontal components. The vertical component regulates vehicle altitude, while the horizontal component generates translational and yaw motions. This configuration enables the same propulsion system to perform both lift and maneuvering functions without additional actuators, thereby reducing the overall system mass and mechanical complexity.

II-D. Electronic Architecture and Power Stage

The embedded control system is built around an ESP32-C6 Mini microcontroller. This device was selected because it integrates 2.4,GHz Wi-Fi connectivity, eliminating the need for an external communication module, while its RISC-V architecture and hardware LEDC peripheral provide independent PWM generation for the propulsion motors and the vectoring servo with negligible CPU overhead. This allows the processor to simultaneously execute the control algorithms and the onboard web server while remaining compatible with the platform's strict mass constraints.

The platform is powered by a single-cell (1S, 3.7 V nominal) 1000 mAh LiPo battery. A single-cell configuration minimizes system mass while providing sufficient discharge capability to supply the peak current demanded by the coreless DC motors. The battery capacity represents a compromise between flight endurance and payload weight, taking advantage of the available lifting margin provided by the 50-inch helium balloon. The SG90 servo is supplied from a regulated 5 V rail, whereas the ESP32-C6 and all sensors operate at 3.3 V.

Each propulsion motor is driven by an independent PWM-controlled MOSFET power stage. This configuration provides electrical isolation between the inductive motor loads and the microcontroller while enabling efficient high-frequency switching and reliable current handling.

II-E. Altitude and Inertial Sensing

The sensing architecture relies on two sensors connected through a shared I^2C bus, providing the signals required for measuring altitude and yaw behaviors.

VL53L1X Time-of-Flight (ToF) laser sensor: This sensor measures the distance to the ground.It operates in continuous ranging mode using the Pololu driver library, which ensures a constant sampling rate while avoiding the overflow issues commonly encountered with generic implementations. The `XSHUT` pin is permanently tied to 3.3 V, keeping the sensor continuously enabled throughout operation. Laser-based ranging provides millimeter-level resolution and is largely insensitive to ambient illumination, making it well suited for indoor altitude estimation. The sensor provides a reliable indoor measurement range of approximately 4 m, adequate for the ceiling height available for experimental validation.

MPU6050 Inertial Measurement Unit (IMU): The IMU provides three-axis angular velocity and linear acceleration measurements. The yaw rate, r, is integrated to estimate the vehicle orientation on the z-axis, ψ.

III. System Dynamics and Parameter Identification

Since the architecture of the blimp allows separated motion control for translation and rotation on the vertical axis, we decided to model those motions and identify the parameters of such models.

III-A. Vertical Dynamics

The vertical dynamics of the proposed VTOL blimp are governed by the balance between buoyancy, weight, propulsion, and aerodynamic drag. Applying Newton's second law along the vertical axis yields

$$m\ddot{z} = F_T + F_B - mg - F_D,$$

where m is the total vehicle mass, z denotes the altitude, F_T is the thrust generated by the propulsion system, F_B is the buoyant force produced by the helium envelope, and F_D represents the aerodynamic drag.

Prior to the identification experiments, the vehicle was mechanically trimmed until a quasi-neutral buoyancy condition was achieved. Small pieces of modeling clay were added to the gondola so that the buoyant force approximately balanced the vehicle weight,

$$F_B \approx mg.$$

Under this operating condition, the propulsion system is only required to compensate for the small residual imbalance and to generate the thrust required for vertical acceleration. Consequently, the vertical dynamics simplify to

$$m\ddot{z} = F_T - F_D.$$

For the relatively low operating velocities considered in this work, the aerodynamic drag is approximated as a viscous force,

$$F_D = b\dot{z},$$

where b is an equivalent damping coefficient. Furthermore, the thrust generated by the pair of vertically oriented propellers is assumed to be proportional to the applied control input,

$$F_T = k_u\, u,$$

where u denotes the PWM command and k_u is the equivalent thrust constant. Combining these assumptions yields

$$m\ddot{z} + b\dot{z} = k_u\, u.$$

Taking the Laplace transform with zero initial conditions gives

$$\frac{Z(s)}{U(s)} = \frac{k_u}{ms^2 + bs} = \frac{K}{s(\tau s + 1)}, \tag{2}$$

where

$$K = \frac{k_u}{b}, \qquad \tau = \frac{m}{b}.$$

The resulting transfer function corresponds to a second-order system with an integrator, reflecting the fact that altitude is obtained by integrating the vertical velocity.

The parameters K and τ were identified from closed-loop experiments. Open-loop tests were not suitable because the limited 4 m measurement range of the VL53L1X Time-of-Flight sensor could cause the blimp to leave the sensing region or collide with the ceiling before sufficient data were collected. Therefore, a proportional controller with gain $K_p = 100$ was employed exclusively to keep the vehicle within the operating region during the identification experiments.

Since the propulsion system exhibits a noticeable dead zone, an offset of 39 PWM counts was added to the control input. This ensured that the motors operated above their activation threshold during the identification tests, reducing the influence of actuator dead-zone nonlinearity on the identified model. The altitude response was acquired at a sampling period of T=0.5 s using the VL53L1X sensor.

For parameter identification, the closed-loop transfer function obtained by combining the proportional controller with the plant model in (2) has the form of a conventional underdamped second-order system with a constant numerator. An initial approximation of the closed-loop model was obtained from the maximum overshoot and peak time of the experimental response, which resulted in a very low damping ratio. A closer agreement with the experimental response was subsequently obtained by estimating the exponential decay rate and damped oscillation frequency directly from the measured response. The open-loop transfer function describing the vertical dynamics of the blimp was then obtained from the identified closed-loop transfer function as

$$G(s) = \frac{0.01311}{17.19s^2 + s}.$$

Fig. 2 compares the experimental response (blue dotted line) with the identified model (red solid line), both obtained in closed loop with $K_p = 100$, for the identification data obtaining a 83.76 % of fitting. Whereas for the validation data (not shown) we obtained a 78.43 % fit. As expected, the linear approximation does not fully capture the vertical dynamics of the blimp. The main discrepancies are observed in the overshoot and in the small residual oscillations present in the experimental response, which are not reproduced by the identified model. These effects could potentially be captured by a more complex model. Despite these discrepancies, the

identified model provides a sufficiently accurate approximation of the dominant vertical dynamics and can therefore be used for controller design and analysis.

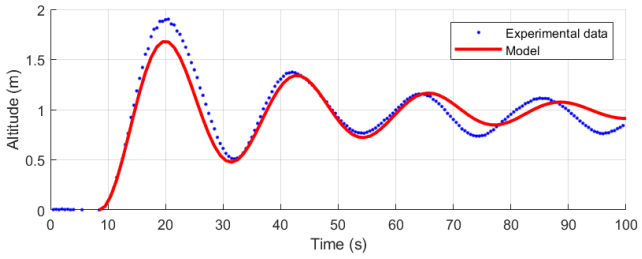


Figura 2. Vertical motion parameter identification experiment.

III-B. Yaw Dynamics

The yaw motion is generated by the differential thrust produced by the propulsion system when the thrust vector is oriented along the horizontal axis. Since the propulsion motors are mounted symmetrically with respect to the vehicle centerline, the applied thrust generates a control torque about the vertical axis, while the surrounding air produces an opposing aerodynamic drag torque. Applying Newton's second law for rotational motion about the vertical axis yields

$$J_z \dot{r} = M_T - M_D, \tag{3}$$

where J_z is the moment of inertia about the vertical axis, M_T is the control torque generated by the propulsion system, and M_D is the aerodynamic resisting torque.

The aerodynamic drag is modeled as the sum of viscous and quadratic components,

$$M_D = b\, r + d\, r|r|, \tag{4}$$

where the linear term dominates at low angular velocities, whereas the quadratic term accounts for the nonlinear aerodynamic effects that become increasingly significant as the yaw rate increases.

Since the linearized yaw dynamics have the same mathematical structure as the vertical dynamics, similar closed-loop behavior was initially expected. However, experimental results revealed a pronounced asymmetry between clockwise and counterclockwise rotations, preventing satisfactory controller performance when identical actuator gains were assumed. This asymmetry is attributed to manufacturing tolerances, differences in propeller efficiency, and minor mechanical misalignments between the two propulsion units.

Accordingly, the yaw dynamics are represented by a piecewise-linear actuator model combined with nonlinear aerodynamic damping. The control torque is therefore modeled as

$$M_T = \begin{cases} k_+ u, & u \geq 0, \\ k_- u, & u < 0, \end{cases} \tag{5}$$

where u is the differential PWM command applied to the propulsion system, i.e. $u = u_l - u_r$.. Positive values of u activate the left motor, whereas negative values activate the right motor.

Substituting (4) and (5) into (3), and dividing by J_z, yields

$$\dot{r} = \begin{cases} a_+\, u - \beta\, r - \gamma\, r|r|, & u \geq 0, \\ a_-\, u - \beta\, r - \gamma\, r|r|, & u < 0 \end{cases}$$

with $a_+ = \dfrac{k_+}{J_z}$, $a_- = \dfrac{k_-}{J_z}$, $\beta = \dfrac{b}{J_z}$, $\gamma = \dfrac{d}{J_z}$.

Unlike the altitude dynamics, the yaw dynamics were identified from open-loop experiments because the rotational motion does not compromise the operating range of the onboard sensors. Constant positive and negative PWM commands covering the operating range of interest were applied for 30 s. This excitation duration allowed the yaw rate to reach steady state before the input was removed, ensuring that both the transient response and the steady-state angular velocity were captured.

Yaw rate measurements were obtained from the MPU6050 gyroscope after gyroscope drift compensation, while the yaw angle was computed by numerical integration of the measured angular velocity.

The model parameters $a_+ = 0.0041$, $a_- = 0.0017$, $\beta = 0.0275$, and $\gamma = 0.0491$ were estimated by nonlinear least-squares optimization in MATLAB, minimizing the error between the measured and simulated yaw responses. The identified model accurately reproduces both the transient response and the steady-state angular velocity over the operating range considered for controller design for the identification data in a 93.77 % of fitting, as illustrated in Fig. 3. Whereas for the validation data (not shown) the coincidence obtained was of 88.73 % of the fitting. The resulting piecewise nonlinear model provides a compact yet accurate representation of the yaw dynamics while explicitly accounting for both actuator asymmetry and nonlinear aerodynamic damping.

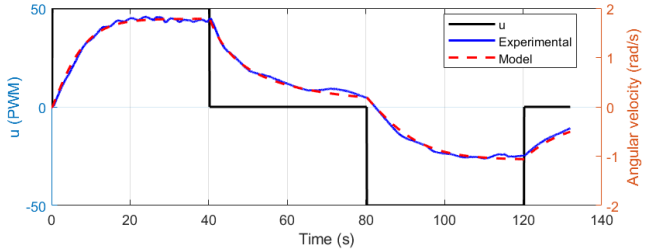


Figura 3. Yaw model parameter identification experiment.

IV. Experimental Implementation and Operational Considerations

The experimental deployment of underactuated platforms based on aerostatic lift presents logistical and physical challenges that differ from those of conventional multirotors. Based on the experimental experience with the proposed blimp, two operational considerations were found to be particularly important for ensuring the integrity of the system and the repeatability of the experimental tests.

Quasi-neutral buoyancy before each flight: It is essential to ensure that the vehicle is close to a state of neutral

buoyancy immediately before each experimental session, in order to avoid the actuators to continuously compensate for its weight and, therefore, extend its autonomy. Since the buoyant force gradually changes due to helium permeation, the vehicle should be trimmed before each experimental session using small amounts of malleable ballast, such as modeling clay, until it exhibits negligible vertical motion in the absence of propulsion input.

Large spatial requirements for dynamic characterization: Experimental identification of the vehicle dynamics requires the application of excitation signals over sufficiently long time intervals. Due to the relatively slow dynamics of the blimp and its large cross-sectional area, considerable time and displacement may be required to observe its characteristic dynamic response. When these experiments are conducted in confined laboratory spaces, the available displacement may be insufficient to complete the excitation maneuvers without risking collisions with surrounding objects. In addition, proximity to walls and other obstacles may introduce aerodynamic disturbances that affect the experimental measurements. Therefore, dynamic characterization should preferably be performed in large, unobstructed, and relatively quiescent environments, such as gymnasiums, indoor halls, or hangars.

V. Conclusions and Future Work

V-A. Conclusions

The design and hardware implementation of the proposed VTOL blimp demonstrated the feasibility of a lightweight aerial robotics platform based on quasi-neutral buoyancy and minimal sensing. The resulting system integrates a 50-inch Mylar balloon, an ESP32-C6-based architecture, a rigid inelastic suspension, and a 180° vectored coreless propulsion system, providing a simple and mechanically stable experimental platform.

Experimental characterization showed that the main vertical and yaw dynamics can be represented by relatively simple models under low-speed operating conditions. The identified models captured the dominant behavior observed experimentally and provide a basis for further analysis and control development. The experiments also highlighted practical limitations of the minimal sensing configuration, particularly the long-term drift of yaw estimation from the MPU6050 gyroscope and the influence of actuator dead zones on experimental identification.

Finally, repeatable characterization requires careful adjustment of the quasi-neutral buoyancy condition and sufficiently large, unobstructed experimental spaces. These results establish the proposed blimp as a simple and accessible platform for aerial robotics research and provide a basis for future extensions involving improved sensing, state estimation, and autonomous operation.

V-B. Future Work

The experimental results suggest three main directions for future development of the platform. First, the available payload could be increased by improving the aerostatic lift, for example through the use of higher-purity helium. This would provide additional margin for future hardware and sensing upgrades while maintaining the current envelope.

Second, an alternative propulsion architecture could be investigated to reduce the coupling introduced by the current thrust-vectoring mechanism. In particular, when helium losses become significant, part of the available propulsive effort must be allocated to compensate for the resulting change in vertical lift, reducing the authority available for longitudinal motion. A propulsion configuration with dedicated actuation for vertical and longitudinal thrust could therefore provide greater flexibility as the buoyancy conditions of the vehicle change.

Finally, feedback controllers could be designed and experimentally evaluated using the identified vertical and yaw models. Long-duration closed-loop experiments would allow the reliability and stability of the current sensing configuration to be assessed over time, providing experimental evidence on whether additional or alternative sensors are required for more reliable altitude and orientation measurements.

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